

From Signals to Cognition: Self-Supervised Multimodal Learning for Cognitive State Inference from Wearable Sensor Streams

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Abstract

In this work, we present a self-supervised multimodal learning framework for wearable sensor data, designed to infer both physical activities and latent cognitive states in real-world environments, which are difficult to annotate and require label-free inference methods. The system leverages three complementary input modalities: eye-tracking, ambient audio, and inertial motion signals to capture rich behavioral and mental cues. These underutilized modalities particularly gaze and audio, offer unique insights into cognitive dynamics and are crucial for detecting mental states beyond motor activity. Each modality is encoded via specialized networks and aligned in a shared representation space using a combination of contrastive learning, masked signal modeling, and cross-modal prediction. Inspired by recent advances such as MU-MAE, CroSSL, RelCon, and PFML, our architecture is robust to missing modalities and optimized for few-shot cognitive inference tasks. A modular prediction head supports diverse downstream applications, including human activity recognition (HAR), mental fatigue detection, and engagement estimation. Through a systematic review of over 20 recent works, we synthesize a taxonomy of techniques at the intersection of self-supervised learning (SSL), multimodal fusion, and cognitive sensing. Our proposed framework demonstrates resilience to sensor noise, asynchronous data streams, and annotation scarcity, making it a promising direction for generalizable and scalable wearable AI systems. The full implementation of our framework, CognifySSL, is available at: <https://github.com/pachaboul/cognifyssl>

CCS Concepts

• **Computing methodologies** → **Machine learning**; *Natural language processing*; • **Human-centered computing** → **Ubiquitous and mobile computing**; • **Applied computing** → *Health informatics*.

Keywords

Self-Supervised Learning, Multimodal Fusion, Wearable Sensors, Cognitive State Detection, Human Activity Recognition

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1 Introduction

Wearable technologies equipped with motion sensors such as Inertial Measurement Units (IMUs), eye-trackers, and microphones are increasingly deployed to capture human behavior in naturalistic environments such as healthcare, education, and interactive systems. These devices continuously generate high-dimensional, multimodal data streams that hold great promise for understanding not only what people are doing, but also how they are thinking and feeling.

Despite this potential, most existing research remains centered on the recognition of physical activities. These approaches often rely exclusively on accelerometer data and require large amounts of labeled examples, making them difficult to scale and generalize across contexts and users [10].

In comparison, cognitive states such as attention, stress, and mental fatigue are still largely overlooked, even though they are critical indicators in learning and well-being. This gap is particularly striking given recent advances in self-supervised learning (SSL)[20], which allow models to learn useful representations from unlabeled data, and early findings suggest that modalities like gaze and audio can reveal meaningful patterns related to internal mental processes [4, 18].

This poster proposes a shift in focus: leveraging SSL and multimodal sensor fusion as a pathway to infer latent cognitive states in a label-free and robust manner.

2 Motivation & Problem Statement

Real-world sensing introduces unique challenges not present in controlled lab environments.

First, cognitive states such as attention, fatigue, or stress are inherently subjective, temporally dynamic, and difficult to annotate. Unlike observable physical actions (e.g., walking), mental states typically require costly or intrusive labeling methods (e.g., questionnaires, clinical assessments), which are unsuitable for large-scale or long-term deployment. As a result, traditional supervised learning approaches are often impractical for cognitive modeling tasks [13].

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Second, real-world wearable data is noisy, misaligned, and multimodal, with frequent sensor dropouts and artifacts (e.g., gaze occlusion, audio noise). These challenges hinder robust fusion across heterogeneous platforms [4, 7].

Third, existing work often isolates SSL, multimodal fusion, and cognitive state modeling. Despite SSL's success in HAR [20] and advances in fusion, few frameworks integrate both for cognitive inference or handle missing modalities with adaptive attention.

There is a pressing need for architectures capable of learning from weak, unlabeled, and multimodal sensor data. This work addresses that gap by reviewing recent SSL and fusion advances and proposing a scalable, label-free framework for cognitive sensing.

No single modality sufficiently captures mental states. IMUs detect motion but miss attention or stress; gaze offers attention cues but is error-prone; audio captures speech context but is noise-sensitive. Only through multimodal fusion can cognitive states be inferred reliably—yet this remains challenging due to signal misalignment and missing data [4, 7, 11].

This work proposes a conceptual framework that unifies self-supervised learning, cognitive state modeling, and multimodal fusion (gaze, audio, IMU) with built-in robustness to missing data. Drawing from prior conceptual designs such as SOAR [7], CroSSL [4], and EyeKnowYou [13], it addresses the key gaps summarized in Table 2, and serves as a blueprint for label-efficient cognitive AI.

3 Method (Systematic Review Summary)

To inform our framework design, we systematically reviewed recent literature on SSL, multimodal fusion, and cognitive inference.

To establish a foundation for our proposed framework, we conducted a systematic review of recent research at the intersection of self-supervised learning (SSL), multimodal sensor fusion, and cognitive or affective state detection. Our goal was to identify existing trends, challenges, and methodological gaps in the literature, as well as to synthesize a taxonomy that informs future development.

3.1 Data Sources and Search Strategy

We searched four major repositories of scientific literature: IEEE Xplore, ACM Digital Library, arXiv, and PubMed, covering the period from 2020 to 2025. Our query strategy focused on terms including “self-supervised learning,” “multimodal fusion,” “wearable sensors,” “cognitive state detection,” and “affective computing.”

3.2 Inclusion Criteria

For inclusion, a paper needed to satisfy at least two of the following three criteria:

- (1) It introduced or evaluated self-supervised learning techniques (e.g., contrastive learning, masked modeling, cross-modal prediction) for time-series or multimodal data [4, 20].
- (2) It involved fusion of two or more modalities, such as IMU, gaze, audio, EEG, or physiological signals [10, 18].
- (3) It addressed cognitive or affective state recognition, such as attention, fatigue, engagement, or emotional stress [7, 13].

3.3 Coding Dimensions

Each selected study was manually coded according to the following analytical dimensions:

- **Modalities used:** including IMU (accelerometers, gyroscopes), audio, eye-tracking, EEG, and ambient context sensors.
- **SSL techniques:** such as contrastive methods (e.g., SimCLR, BYOL), masked autoencoders (e.g., MAE, MU-MAE), and temporal prediction [9, 16].
- **Targeted cognitive/affective states:** including but not limited to stress, focus, mental workload, and emotional arousal.
- **Evaluation protocols:** covering accuracy, F1-score, AUC, cross-modal transfer, few-shot and zero-shot generalization, and robustness to missing modalities [3, 4].

3.4 Output: Taxonomy and Gap Analysis

The outcome of this review is a taxonomy that classifies studies across four axes: (1) **modality combinations** (e.g., IMU-only vs. IMU+Gaze+Audio), (2) **SSL paradigm used** (e.g., contrastive vs. masked modeling), (3) **target cognitive state** (e.g., engagement vs. fatigue), and (4) **fusion robustness** (e.g., whether models explicitly handle missing modalities).

For instance, [4] emphasizes cross-modal prediction with robustness to sensor dropout, whereas [18] focuses on unimodal gaze-audio fusion without explicit missing data handling.

This taxonomy highlights recurring gaps: the scarcity of real-time or few-shot evaluation protocols, limited resilience to asynchronous or missing sensor inputs, and underutilization of rich modalities like eye-tracking and ambient audio for cognitive inference. As illustrated in Figure 1, most reviewed studies cluster around IMU-based models for physical activity recognition using contrastive methods, with limited attention to gaze or audio inputs and minimal handling of missing modalities. This visual synthesis helps identify underexplored combinations and motivates the need for more robust multimodal architectures.

This systematic synthesis serves as a roadmap for designing scalable, label-free architectures that move beyond physical activity recognition toward deeper models of human cognition.

4 Findings & Gap Analysis

The systematic review revealed several important patterns and limitations in current research at the intersection of self-supervised learning (SSL), multimodal sensing, and cognitive state detection.

First, the vast majority of reviewed studies continue to focus on physical human activity recognition (HAR). While SSL has been effectively used to improve HAR tasks from IMU signals [20], cognitive states such as attention, mental fatigue, or emotional stress remain largely underrepresented in these applications.

Second, although multimodal sensing has become more accessible, many works still rely on unimodal data, typically from IMU or biosignals like heart rate or skin conductance. Gaze and audio modalities, which are strongly linked to internal cognitive and affective processes [13, 18], are rarely used or properly fused.

Third, while self-supervised learning methods are widely applied, they are often developed and tested under ideal or complete-modality conditions. Very few systems implement fusion strategies that are robust to missing or corrupted data streams an essential requirement for real-world deployment [4, 7].

Finally, only a small subset of studies attempt to decode or enhance cognitive states like stress, fatigue, or attention fluctuations.

Table 1: Taxonomy of Selected Multimodal SSL Studies

Study references	Modalities Used	SSL Paradigm	Target Cognitive State	Fusion Robustness
Deldari et al. (2024) [4]	IMU + Audio + Gaze	Cross-modal Prediction	Fatigue, HAR	Handles missing modalities
Wu et al. (2023) [18]	Gaze + Audio	Masked Modeling	Emotion	Not addressed
Das et al. (2024) [3]	IMU	Masked Modeling	Activity	Partial handling
Xu et al. (2024) [19]	IMU	Contrastive Learning	Activity	Robust to misalignment
Fuhl et al. (2021) [6]	Gaze	Contrastive Learning	Attention	Not addressed

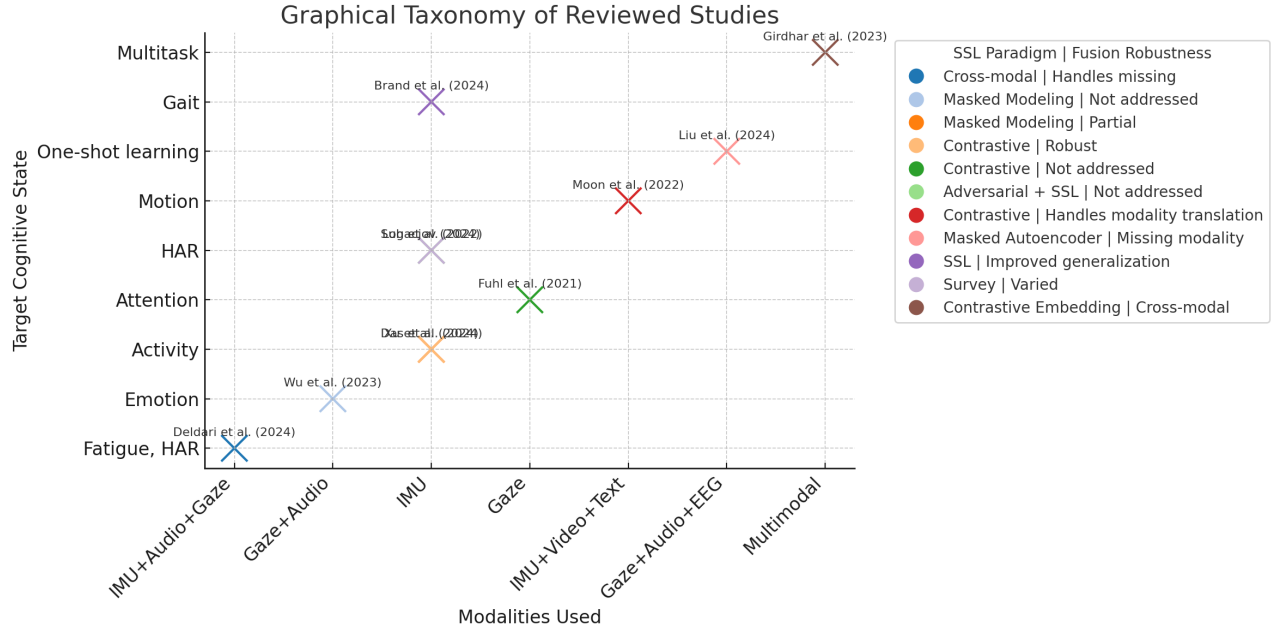


Figure 1: Graphical taxonomy of reviewed multimodal self-supervised learning studies. The horizontal axis shows the modalities used (e.g., IMU, Audio, Gaze), while the vertical axis indicates the targeted cognitive states (e.g., fatigue, emotion, attention). Colors represent SSL paradigms (e.g., contrastive, masked modeling), and hatch patterns encode robustness to missing modalities.

Among these, none simultaneously leverage SSL, full multimodal fusion, and resilience to sensor unreliability that highlighting a crucial opportunity for innovation in wearable cognitive analytics.

These gaps directly informed the design of our framework in Section 5. Specifically, the lack of robust fusion architectures, limited use of underexplored modalities (e.g., gaze and audio), and the scarcity of cognitive-focused models inspired our modular SSL-based design.

These limitations motivated the development of a unified, modular architecture that aligns multi-modal inputs using SSL, while ensuring resilience to missing modalities and cognitive signal ambiguity.

5 Proposed Architecture

The proposed architecture is directly informed by the gaps identified in Sections 3 and 4, notably the absence of cognitive state modeling, weak robustness to missing modalities, and underuse of gaze and

audio signals. Each module is intentionally designed to overcome these limitations. Table 2 details the alignment between observed shortcomings and our architectural choices.

We now describe the design of our proposed multi-modal SSL framework for wearable cognition-aware inference.

We propose a modular, self-supervised architecture designed for robust inference of both physical activities and latent cognitive states using heterogeneous multi-modal data. The framework is structured to maximize flexibility, extensibility, and resilience to missing or degraded modalities.

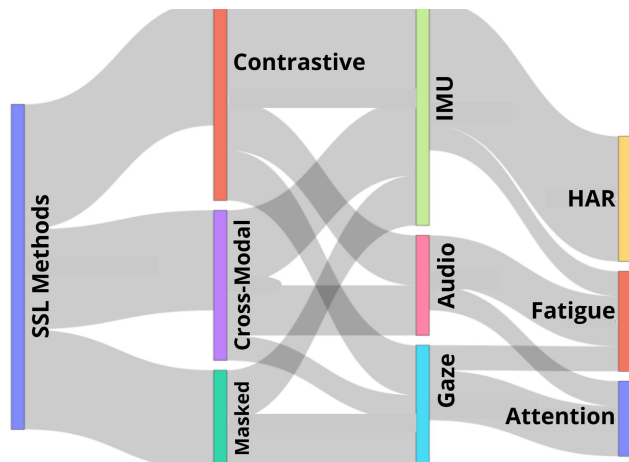
5.1 Input Modalities

The system ingests synchronized streams from three primary sensor sources:

- **Eye-tracking data:** gaze position, blink rate, pupil dilation,
- **Ambient audio signals:** speech, noise, silence, prosody,

Table 2: Mapping Literature Gaps to Architecture Components

Gap Identified in Literature	Corresponding Design Decision
Lack of cognitive state modeling in SSL-HAR studies	Introduce cognitive tasks (fatigue, engagement) in multi-head prediction module
Underutilization of gaze and audio in multimodal setups	Incorporate GazeNet and WaveEncoder as primary encoders
Fusion not robust to missing/occluded modalities	Use dropout-inspired masking + cross-modal prediction losses
Over-reliance on contrastive-only SSL	Combine contrastive, masked modeling, and cross-modal prediction
Lack of real-time/few-shot adaptability	Use modular, shared embedding + few-shot prediction heads


Figure 2: Cross-modal self-supervised learning interactions. The Sankey diagram shows how gaze, audio, and IMU encoders align via SSL tasks to support downstream inference of fatigue, HAR, and engagement.

- **Inertial motion data from IMUs:** accelerometers and gyroscopes.

Each of these modalities captures a complementary aspect of user behavior and cognitive engagement.

5.2 Modality-Specific Encoders

Each input stream is processed by a dedicated encoder module, optimized for its specific data distribution and temporal structure:

- **GazeNet:** a CNN-based encoder for eye movement sequences,
- **WaveEncoder:** a 1D-CNN for raw audio or spectrogram features,
- **IMU Encoder:** a Transformer-based encoder for sequential motion data.

These encoders learn modality-specific embeddings and are pre-trained using pretext tasks adapted to each signal type.

5.3 Shared SSL Embedding Space

All encoder outputs are projected into a shared latent space trained using self-supervised learning objectives:

- **Contrastive learning:** aligning semantically similar samples across and within modalities (e.g., SimCLR, BYOL),
- **Masked signal modeling:** masking parts of the input (e.g., time steps, channels) and reconstructing them, inspired by MAE and MU-MAE,
- **Cross-modal prediction:** predicting missing modality features (e.g., predicting audio from gaze and IMU) [4].

This embedding space enables cross-modality alignment and acts as a robust intermediary between raw signals and high-level inference.

5.4 Multi-Head Prediction Module

From the shared representation, the architecture branches into multiple task-specific output heads, enabling both classification and regression:

- **HAR head:** human activity recognition,
- **Fatigue detection head:** for identifying mental exhaustion,
- **Engagement estimation head:** attention or engagement, either binary or continuous.

Each head can be trained with minimal supervision, fine-tuned if labeled cognitive annotations are available, or evaluated in zero-shot setups.

5.5 Robustness to Missing Modalities

To mimic real-world conditions where sensors may fail or be partially occluded, the architecture includes dropout-inspired training. During training, one or more modalities are randomly masked, and the model is required to reconstruct or predict target features using the remaining inputs enhancing robustness at inference time [8].

6 Conclusion and Future Work

This work proposes a robust, self-supervised multimodal architecture to infer latent cognitive states in real-world settings. By integrating eye-tracking [6, 13], audio [17], and inertial signals [3, 20] into a shared representation space, the framework addresses key challenges such as label scarcity, missing modalities [4, 7], and asynchronous data streams.

The architecture's modularity supports diverse downstream tasks, including human activity recognition [10], fatigue detection [18], and engagement estimation all with minimal supervision.

Future efforts will include implementing the framework on actual wearable devices and validating it against cognitive ground truth signals such as functional near-infrared spectroscopy (fNIRS) [12, 14], heart rate variability (HRV) [5], and behavioral task performance metrics [13]. Such validation will help bridge the gap between algorithmic inference and real-world cognitive measurement.

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